

Yasah Sai Chandra Borusu

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EDUCATION

George Mason University, College of Engineering & Computing

Fairfax, VA

B.S. in Computer & Information Sciences

Aug 2022 – Dec 2026

Relevant coursework: Data Structures & Algorithms, Database Systems, Operating Systems, Machine Learning, Computer Networks, Software Engineering, Web Development, Probability & Statistics

EXPERIENCE

Software Engineer (Founding Team)

Aug 2025 – Apr 2026

TheCollegeTech

Hybrid

- Co-founded a **4-person** engineering team and shipped **3 production modules** — Learning, Campus, and Placement Management — covering attendance & gradebook, timetables, applications/shortlisting, and notifications.
- Built a reusable **React + TypeScript** component library of **25+** tables, forms, and modals with shared validation and typed API clients, cutting repeated UI work across all three modules.
- Designed **30+ Python** REST endpoints over relational **SQL** models with filtering, pagination, role-based access, parameterized queries, and schema migrations; authored unit/integration tests and reviewed teammates' PRs.

Cloud & AI Intern

May 2025 – Jul 2025

Astrani Technology Solutions

Ashburn, VA (Hybrid)

- Built **AI-driven business insights** tools on **AWS Bedrock** over **S3/DynamoDB** datasets; benchmarked **OpenAI**, **DeepSeek**, and **Mistral** for CX analytics, summarization, and recommendations.
- Led **data engineering** migration from on-prem to **AWS RDS (PostgreSQL)**, achieving nearly **70%** infrastructure cost savings; designed ETL workflows and parameterized SQL for downstream reporting.
- Recommended a domain-specific **small-language-model (SLM)** strategy to cut inference cost versus large LLMs; supported image classification, text processing, and chatbot work with experiment tracking.

Software Engineer Intern

May 2024 – Aug 2024

Rise Consultancy

Remote

- Shipped front-end and back-end features for client-facing web applications in **JavaScript**, **React**, and **HTML/CSS**, working within an agile engineering team across the full development lifecycle.
- Integrated **REST APIs** into responsive UI components and validated request/response contracts with **Postman**; queried and debugged **SQL** data issues and managed changes through **Git** branch and review workflows.

Undergraduate Teaching Assistant (Python)

Aug 2023 – Present

College of Engineering & Computing, George Mason University

Fairfax, VA

- Provided one-on-one debugging and conceptual support to **150+ students per semester** across labs, projects, and coursework, diagnosing root causes of errors and walking students through fixes until they could apply the concept independently.
- Served as a recurring point of contact across **6+ semesters** (Aug 2023–Present) for students stuck on Python fundamentals, project logic, and lab assignments, translating instructor material into explanations tailored to each student's specific gap in understanding.

TECHNICAL SKILLS

Languages: Python, SQL (MySQL, PostgreSQL), Java, C/C++, TypeScript/JavaScript

Frameworks & Libraries: React, Next.js, Node.js, FastAPI, LangChain, PyTorch, OpenCV

Cloud & Databases: GCP, AWS (Bedrock, S3, DynamoDB, RDS), Microsoft Azure (Fundamentals), Supabase

Web & Tools: REST APIs, HTML/CSS/XML, Git, Postman, Linux/Unix, MS Office

Certifications: Microsoft Azure Fundamentals (AZ-900), Analytics Vidhya AI/ML Bootcamp

PROJECTS

BRIK-E: Smart-Building Hazard Detection — *Python, PyTorch, YOLOv8/SegFormer, OpenCV, FastAPI*

- Trained and deployed custom computer-vision models to detect spills, trash, and lost items on CCTV feeds; added a **two-stage verifier** (detector to classifier) to reduce false positives.
- Calibrated cameras to floor plans via **homography** for real-world (x, y) targets; built a real-time event pipeline over **MQTT** for robot dispatch and live tracking.

MindEase: AI Task Planning Chatbot & Dashboard — *Next.js, TypeScript, Supabase (Postgres + Auth)*

- Built and deployed a chat app converting natural language into prioritized tasks and a daily action plan, with multi-task capture per message, subtask breakdown, and a live workload dashboard.
- Architected a **Next.js + TypeScript** frontend on **Supabase Postgres** with SSR auth, **row-level security** policies, typed database contracts, and REST routes for chat parsing, task CRUD, and summaries.

Automated Meeting Summarizer — *LangChain, Whisper, GPT-4/LLaMA-2, Google Calendar API*

- Built live transcription with **Whisper** and automated extraction of key points, decisions, and action items into structured outputs; integrated **Trello/Asana** and **Google Calendar** APIs for task creation and deadline tracking.